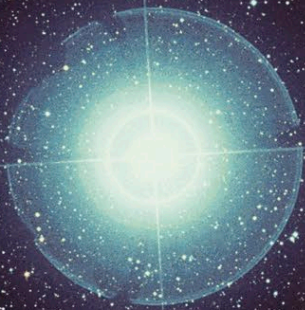




VARIABLE STARS

More than 30,000 variable stars are known within the Milky Way, and it is likely that there are many thousands more waiting to be discovered. Variable star research is a fundamental and vital branch of astronomy, as it provides information about stellar masses, temperatures, structure, and evolution. Variable stars often have periods

ranging from years to decades, and professional astronomers do not have the resources to continuously monitor such stars. Consequently, amateur astronomers play a key role within this field, submitting thousands of observations into an international database.

IRREGULAR VARIABLE
The brightness of the variable star, Gamma (γ) Cassiopeiae, changes irregularly and unpredictably by up to two magnitudes.

ROTATING VARIABLE

Procyon



DISTANCE FROM SUN
11.4 light-years
MAGNITUDE 0.34
SPECTRAL TYPE F5

CANIS MINOR

Procyon is only about seven times as luminous as the Sun but appears bright in the sky due to its proximity

CONSPICUOUS VARIABLE

Seven times more luminous than the Sun, Procyon is the eighth-brightest star in the night sky.

to Earth. Procyon has a companion, Procyon B, a white-dwarf star about the same size as Earth. Procyon shows small changes in magnitude caused by surface features, such as star spots, passing in and out of view as the star rotates. This type of variation classifies Procyon as a BY Draconis-type variable. In addition to surface changes, the tiny, brighter companion also increases the apparent brightness of Procyon when it passes in front of the star as seen from Earth.



ERUPTIVE VARIABLE

U Geminorum



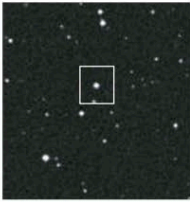
DISTANCE FROM SUN
250 light-years
MAGNITUDE 8.8
SPECTRAL TYPE B

GEMINI

The prototype cataclysmic variable star, U Geminorum, is a close binary system consisting of a red main-sequence star orbiting and eclipsing a white dwarf and its accretion disk. Material falls from the main-sequence star onto the disk, causing localized heating and rapid increases in brightness of three to five magnitudes.

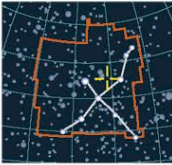
PROTOTYPE

U Geminorum lends its name to a type of irregular variable star that displays sudden increases in brightness.



INDETERMINATE VARIABLE

Tabby's Star



DISTANCE FROM SUN
1,470 light-years
MAGNITUDE 11.7
SPECTRAL TYPE F3V

CYGNUS

The planet-hunting satellite Kepler revealed this faint star (named after codiscoverer Tabetha S. Boyajian) to be one of the most intriguing variables in the sky. Tabby's Star is prone to unpredictable dips that reduce its brightness by up to 22 percent. These dips are too large to be caused by a transiting planet or eclipsing star, but spectroscopic studies show no signs of dust or planet-forming material in orbit. A precise cycle to the eclipses has also proved hard to pin down. One hypothesis is that Tabby's Star is orbited by a swarm of small compact bodies, while others suggest some new type of pulsation mechanism in the star itself.

ECLIPSING BINARY

Eta Geminorum



DISTANCE FROM SUN
349 light-years
MAGNITUDE 3.3
SPECTRAL TYPE M3

GEMINI

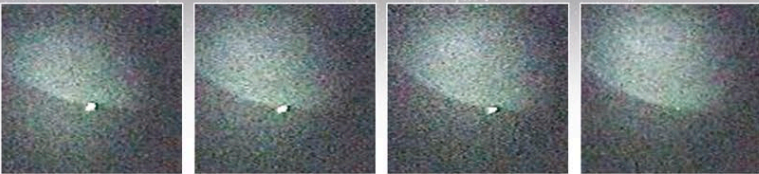
Commonly known as Propus, Eta (η) Geminorum is a red giant star, and its red coloring is very apparent through binoculars. It is a semiregular variable star with a 0.6-magnitude variation—ranging between magnitudes 3.3 and 3.9—over 234 days. Propus is also a spectroscopic eclipsing binary, having a cool spectral-type-B companion star orbiting it with a period of 8.2 years and at a distance of about 625 million miles (1 billion km). Propus is therefore eclipsed every 8.2 years and is a target for amateur variable-star observers. A second star orbits at a greater distance, with a period of 700 years, but with no eclipsing. Although Propus is a cool star, with

CHANCE ALIGNMENT

In Earth's skies, Eta Geminorum lies close to the far more distant supernova remnant IC 433.



a temperature of about 6,500°F (3,600°C), it is more than 2,000 times as luminous as the Sun. Its temperature and luminosity suggest that it is 130 times larger than the Sun. This agrees with optical measurements, but there is some uncertainty—the star has different sizes when observed at different wavelengths due to dark bands of titanium oxide in its spectrum. This uncertainty in measuring Propus's size is typical of large, cool stars. Propus is evolving—observations by amateur



astronomers show that its average brightness has increased by 0.1 magnitude over the last decade. It has a dead helium core and is slowly entering a new phase: it is destined to become a Mira variable (see opposite).

ETA GEMINORUM OCCULTED

In an event that takes less than one-thirtieth of a second (above), Propus is occulted by the Moon (see p.71). In an optical image (top), Propus is pictured alongside the much more distant supernova remnant IC 443.